

X4Pro. News-2026.

by V.Zerkin, Vienna, 2026-09-01

1. Code reac1.py: universal retrieval and plotting of EXFOR data having one-string Reaction-codes, for example:

- CSP Partial cross section data
- CST Temperature dependent cross section
- ETA Average neutron yield per nonelastic event
- NU Fission-neutron yield, nu-bar
- TKE Total kinetic energy
- SIG Cross section data given with limit as DATA and DATA-MIN
- SIG Production cross section, where product coded as SF4 or in DATA (ELEM/MASS)

-- Currently implemented search: by full-Reaction-code, DatasetID, Author1

2. Plotting partial cross sections from local EXFOR and remote ENDF databases

- CSP vs ENDF MT:51/MF:3,33 - Production of a neutron, with residual in the 1st excited state
- NU vs ENDF MT452,455,456/MF:1,31 - Average number of total/delayed/prompt neutrons released per fission event
- ETA vs ENDF MT:452,18,102/MF:1,3,31,33 - Average neutron yield per nonelastic event: $\eta \pm \Delta\eta$

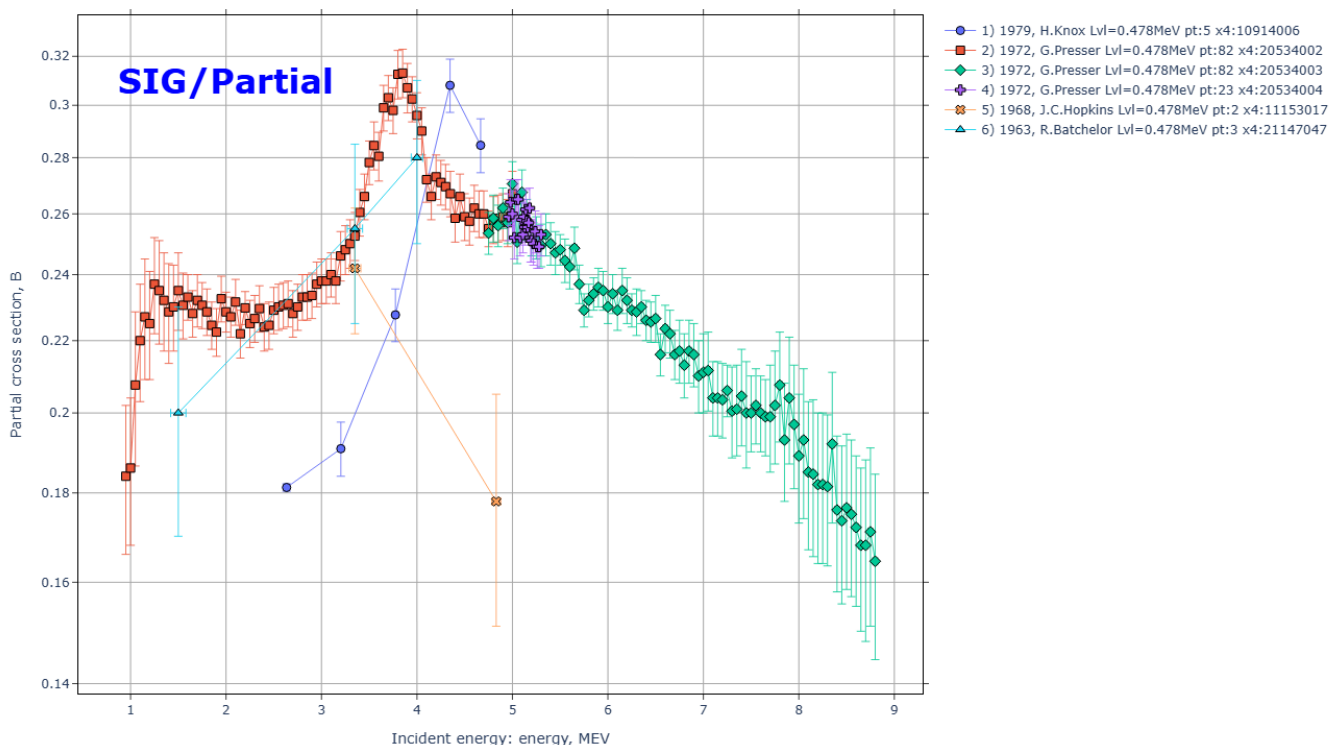
Code reac1.py help:

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Program: reac1.py, ver.2026-08-14
Author: V.Zerkin, Vienna, 2021-2026
Purpose: Retrieve and plot any type EXFOR data from local EXFOR database
Introduction:
* computational data are presented as function y=f(x1,x2,...)
  where f is measured quantity, xi are independent variables:
  EN - incident energy, E2 - energy of outgoing particle,
  ANG - angle of outgoing particle, EVL - energy level, etc.
Algorithm:
* user defines EXFOR Reaction-code for data search, x-variable by number
* selects x-variable by number (default: x1)
* other variables will be used as parameter for grouping of data points
Usage:
$ python [{flag}] x4update.py [{option|reaction}]
* flag: see all Python flags: $ python --help
-B      don't write .pyc files on import
* option:
-help   print this help-text and exit (also --help)
-o:<file> output file name
-x:x?   set x variable: "x1" to "x5", e.g. -x:x2, default: -x:x1
-x?fam:<str> filter for "families" of x? variable, e.g. -x2fam:"LVL;EXC"
-x?min:<num> filter out values of x? variable, e.g. -x1min:6e6
-x?max:<num> filter out values of x? variable, e.g. -x1max:30e6
-nogrp  avoid grouping datasets by reaction-codes on the plot
-fx:<num> set factor for x-values, e.g. -fx:1e6 sets "MeV" on X-Axis
-fy:<num> set factor for y-values, e.g. -fy:1e-3 sets "mb/sr" on Y-Axis
plotting:
-xlog   logarithmic scale of X-Axis
-ylog   logarithmic scale of Y-Axis
-lines  connect points on the plot by lines
-sym    draw border of a symbol of a data point on the plot
-annot:x,y,tst annotation on top of the plot;
        x and y should be given in units of the plot;
        text can include html tags: <b>, <sup>
* reaction:
<reaction> full EXFOR Reaction-code, e.g. "8-0-16(N,EL)8-0-16,,DA"
```

part1-8-reac/ EXFOR universal: SIG, SIG-T, SIG-PAR, DA, DE, DA-PAR, FY, NUBAR, TKE, ETA

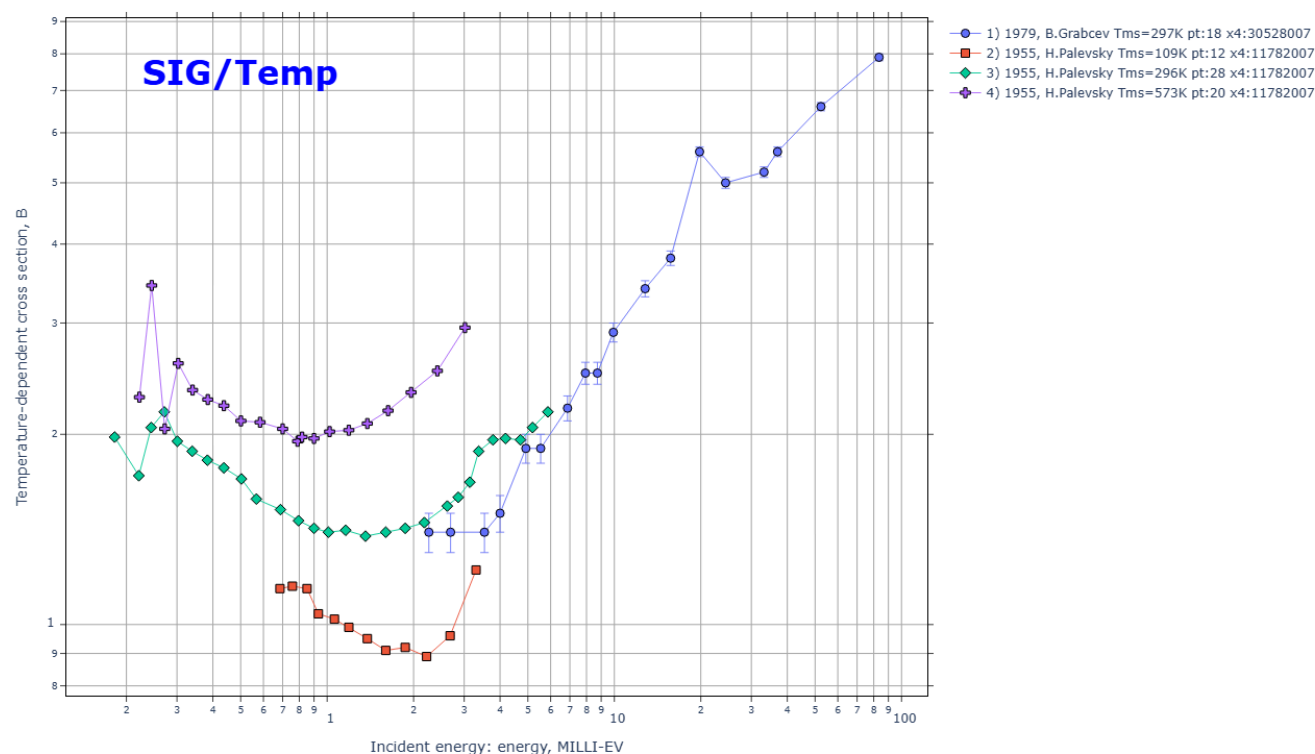
1) part1-8-reac/out00/sig1p.html.png CSP Partial cross section data

Reaction:3-LI-7(N,INL)3-LI-7,PAR,SIG Quantity:CSP:y=Data(En,Lvl) Datasets:6 Points:197
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-08-27 //running:2026-09-01 13:35:01



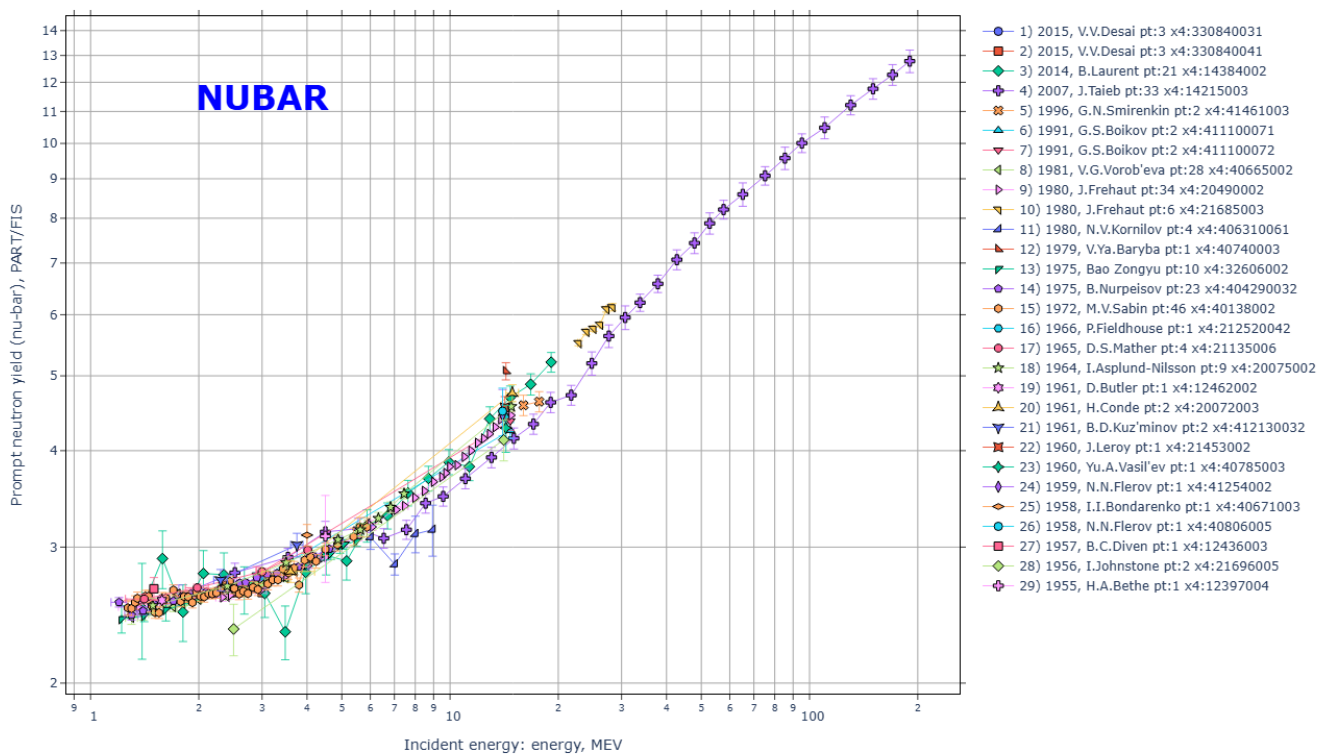
2) part1-8-reac/out00/cst1.html.png CST Temperature dependent cross section

Reaction: 82-PB-0(N,TOT),,SIG/TMP Quantity: CST:y=Data(En,Tms) Datasets: 4 Points: 78
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-08-27 //running:2026-09-01 13:34:50



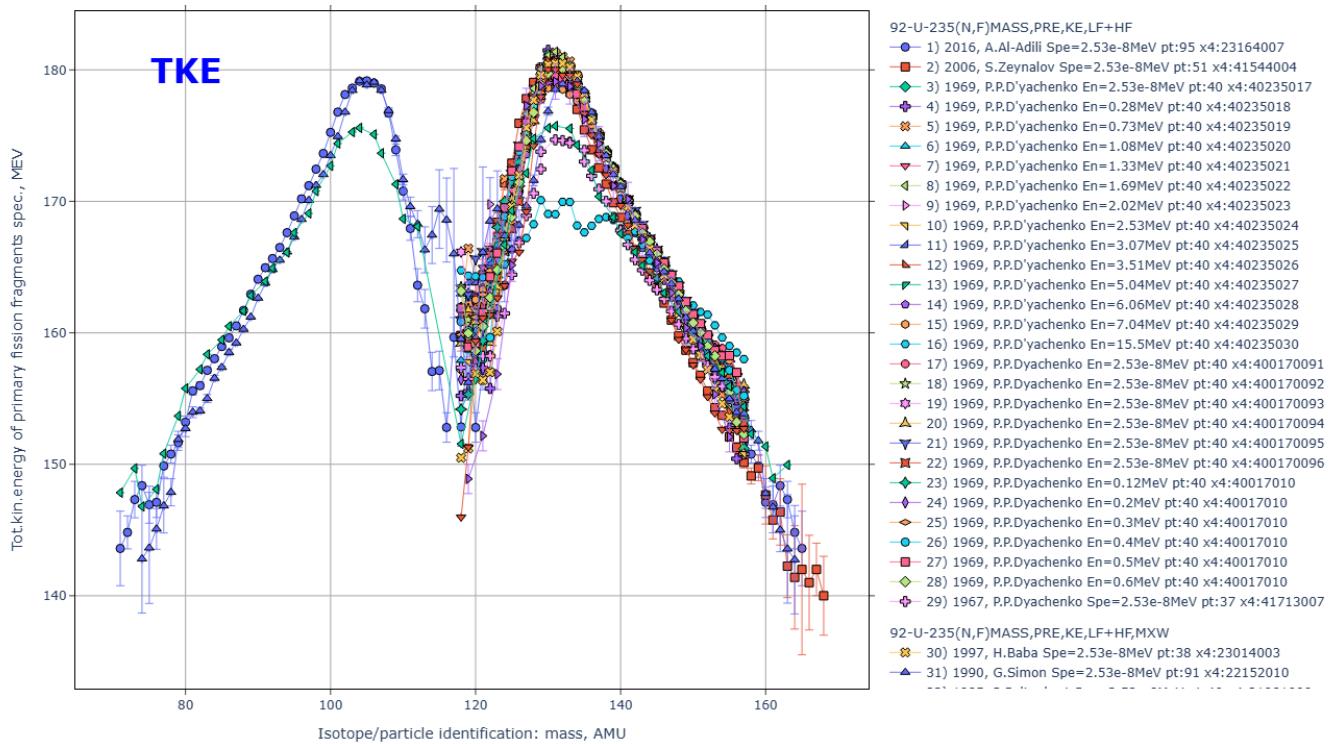
3) part1-8-reac/out00/nu1.html.png NU Fission-neutron yield, nu-bar

Reaction: 92-U-238(N,F),PR,NU Quantity: NU:y=Data(En) Datasets: 29 Points: 246
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-08-27 //running:2026-09-01 13:34:52



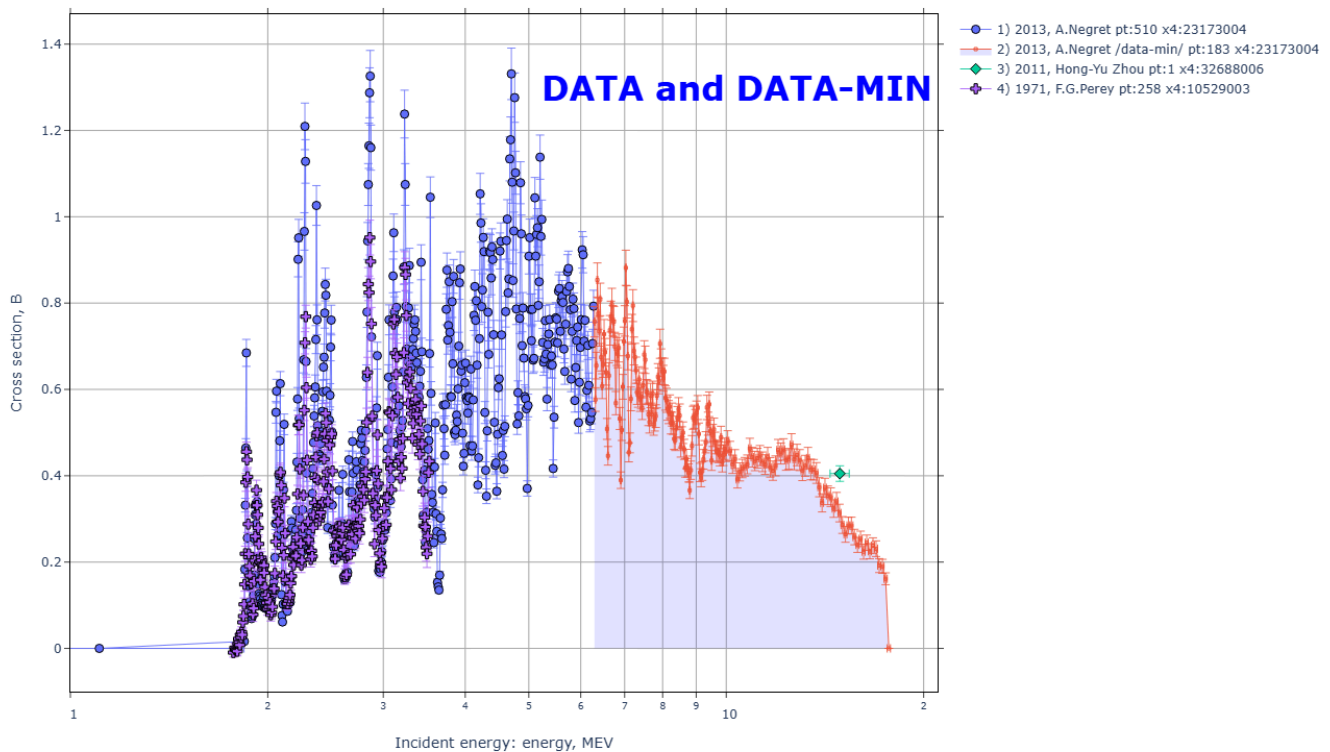
4) [part1-8-reac/out00/tke1.html.png](#) TKE Total kinetic energy

Reaction: 92-U-235(N,F)MASS,PRES,KE,LF+HF Quantity: E: y=Data(Spe,Zam) Datasets: 34 Points: 1467
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-08-27 //running:2026-09-01 13:34:55



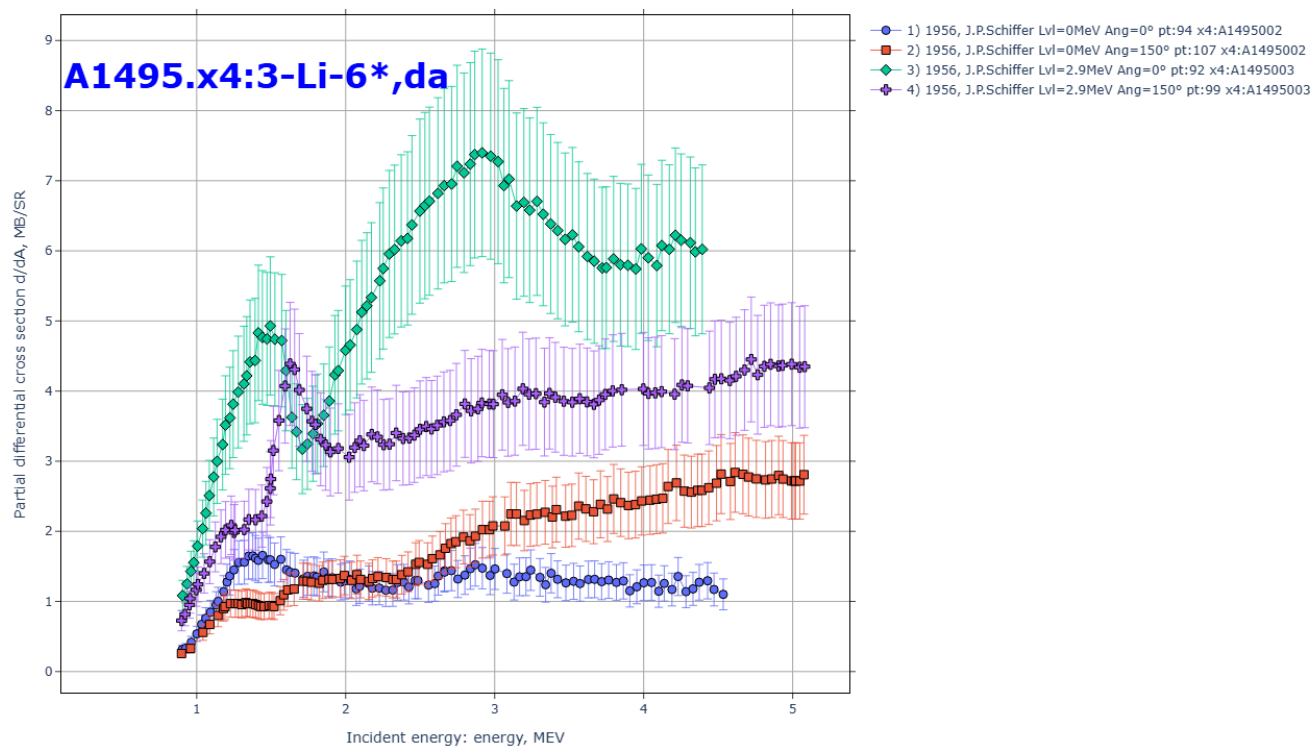
5) [part1-8-reac/out00/si28nn.html.png](#) SIG Cross section data given with limit as DATA and DATA-MIN

Reaction: 14-SI-28(N,INL)14-SI-28,,SIG Quantity: CS: y=Data(En) Datasets: 4 Points: 952
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-08-27 //running:2026-09-01 13:35:03



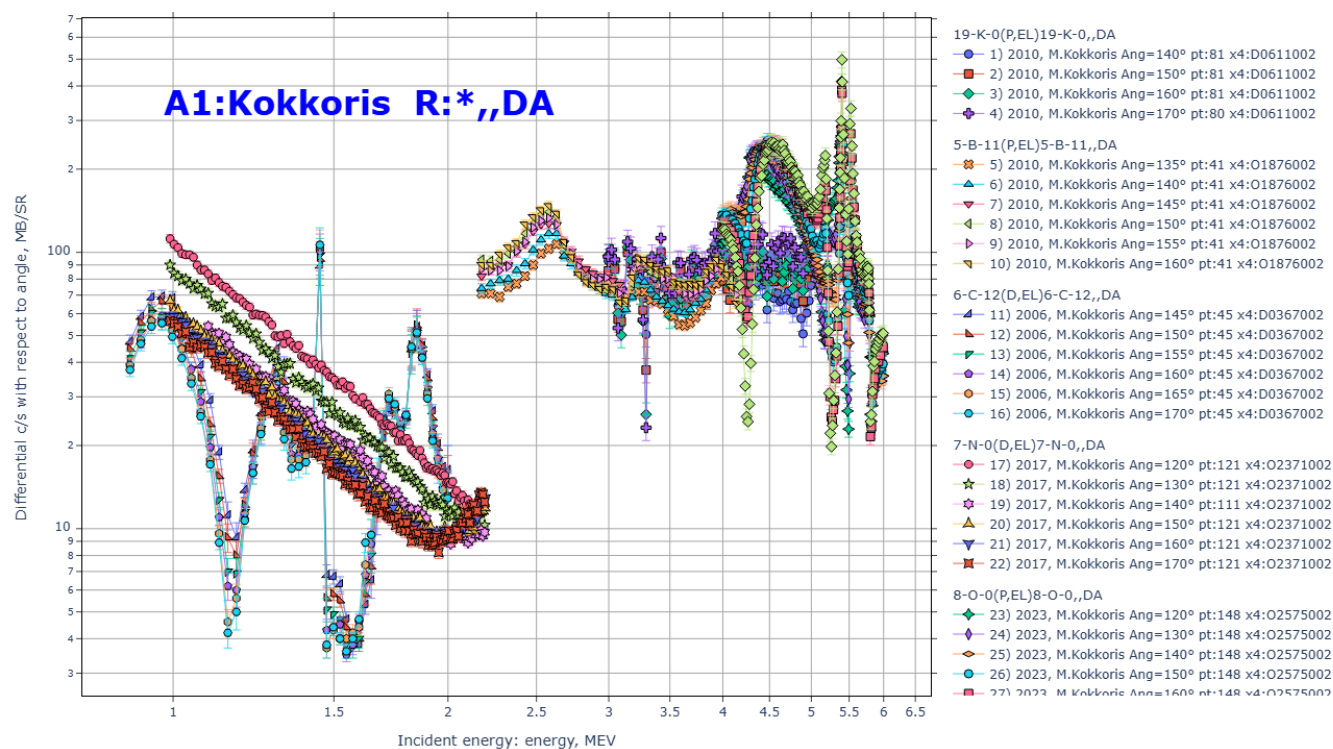
6) part1-8-reac/out00/a1495-li6.html.png Data from EXFOR:"A1495" & Reaction: "3-LI-6*DA"

Reaction: 3-LI-6(HE3,P)4-BE-8,PAR,DA Quantity: DAP:y=Data(En,Lvl,Ang) Datasets: 4 Points: 392
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-08-27 //running:2026-09-01 13:35:06



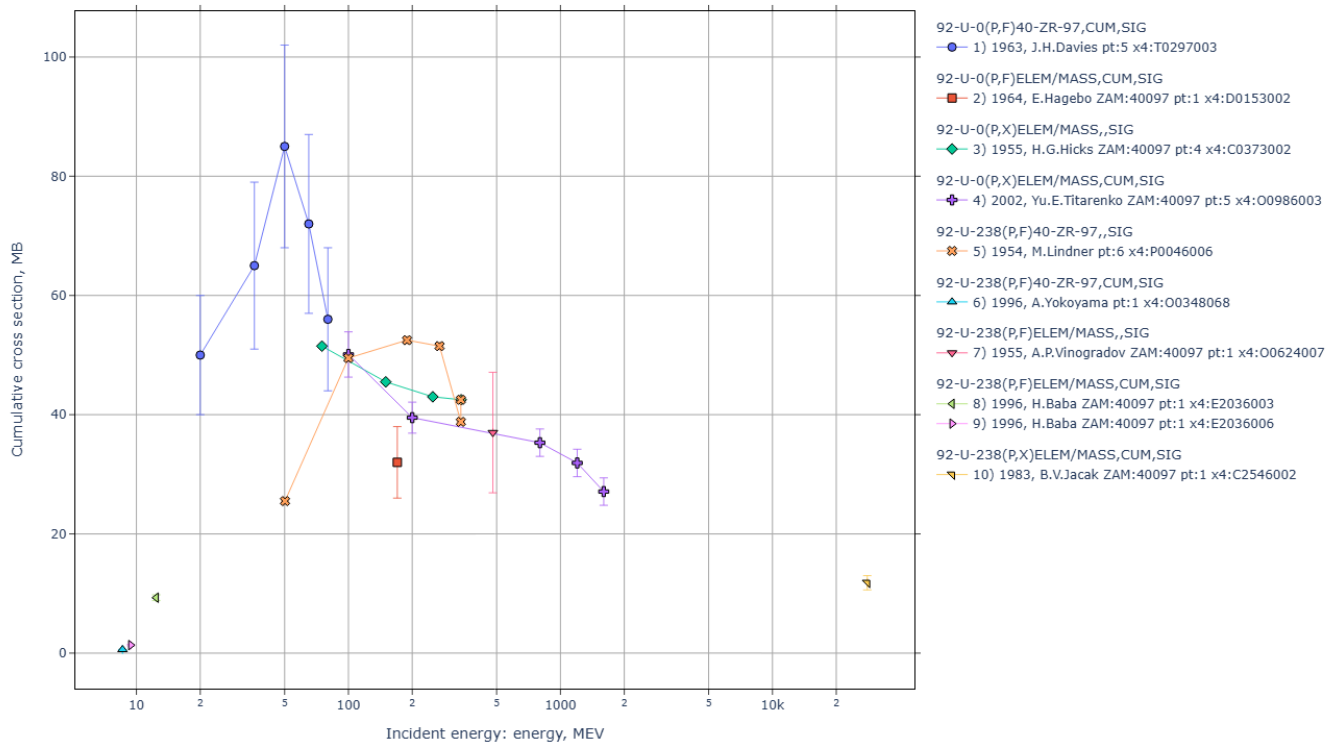
7) part1-8-reac/out00/Kokkoris.html.png Data by "Kokkoris" & Reaction: "*,DA"

Reaction: 19-K-0(P,EL)19-K-0,,DA Quantity: DA:y=Data(En,Ang) Datasets: 28 Points: 2443
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-08-27 //running:2026-09-01 13:35:11



8) part1-8-reac/out00/sig1prod.html.png Production cross section coded as SF4 and ELEM/MASS

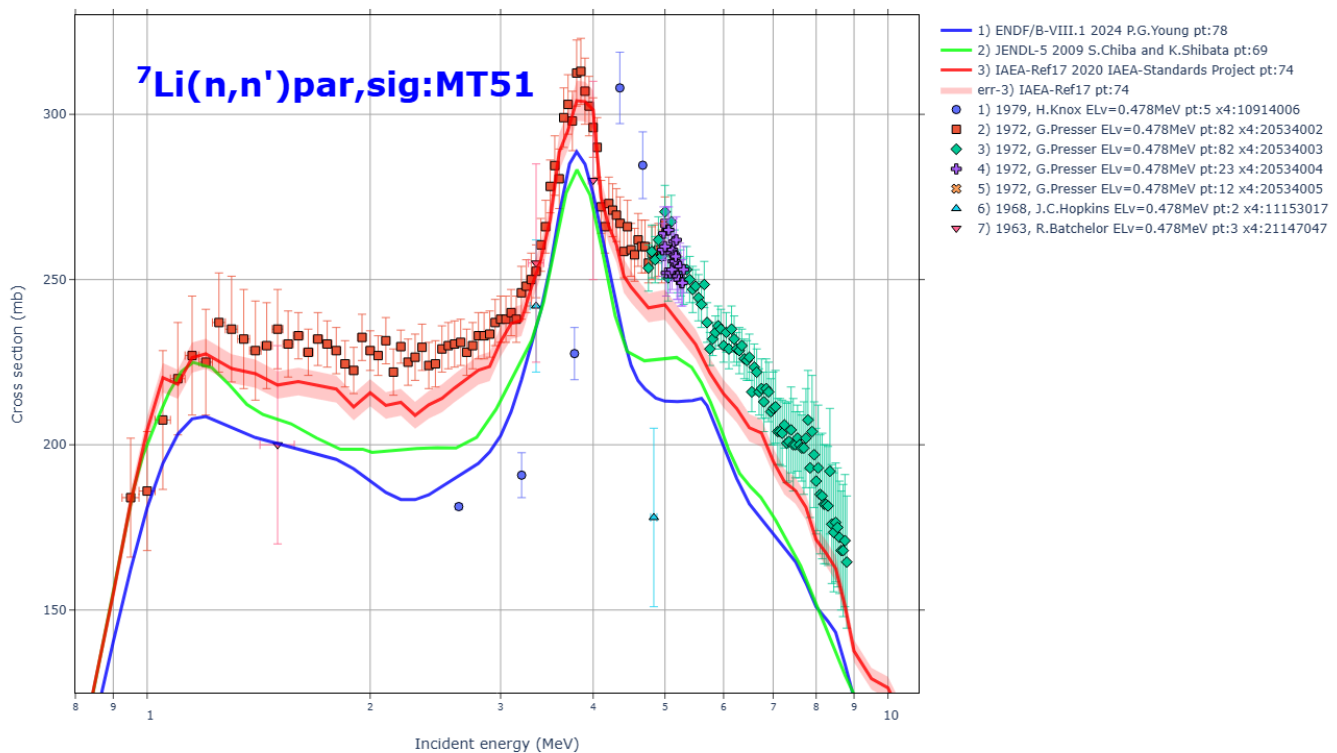
Reaction: 92-U-0(P,F)40-ZR-97,CUM,SIG Quantity:CS:y=Data(En) Datasets:10 Points:26
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-08-27 //running:2026-09-01 13:35:14



part2-8-sig1par/ EXFOR/ENDF partial cross sections CSP:MT51

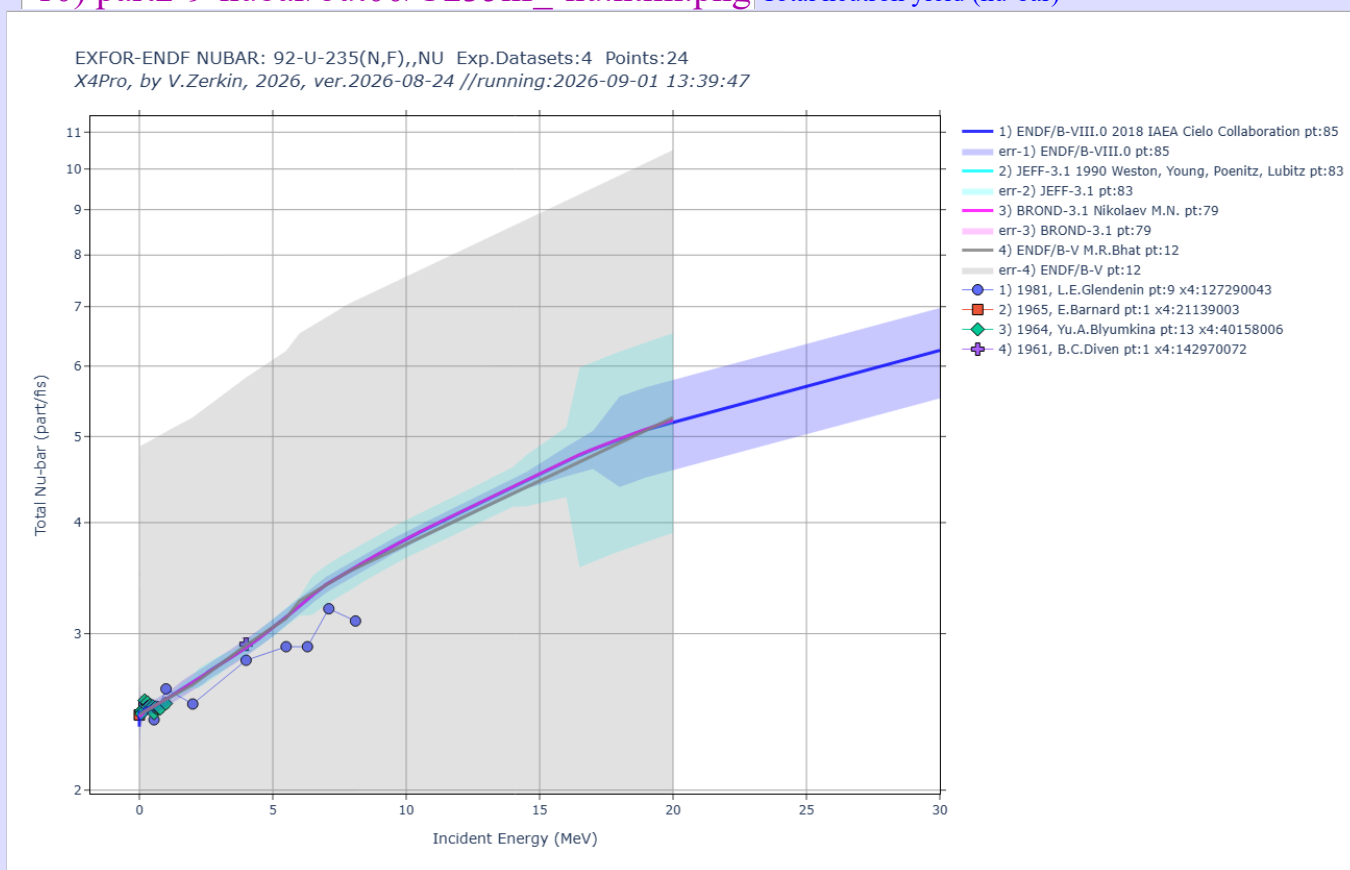
9) part2-8-sig1par/out00/Li7ninl.html.png Production of a neutron, with residual in the 1st excited state

EXFOR/ENDF cross sections SIG(E): Li-7(n,inl)par,sig EXFOR-datasets:7
X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2026, ver.2026-08-10 //run:2026-09-01 13:39:04

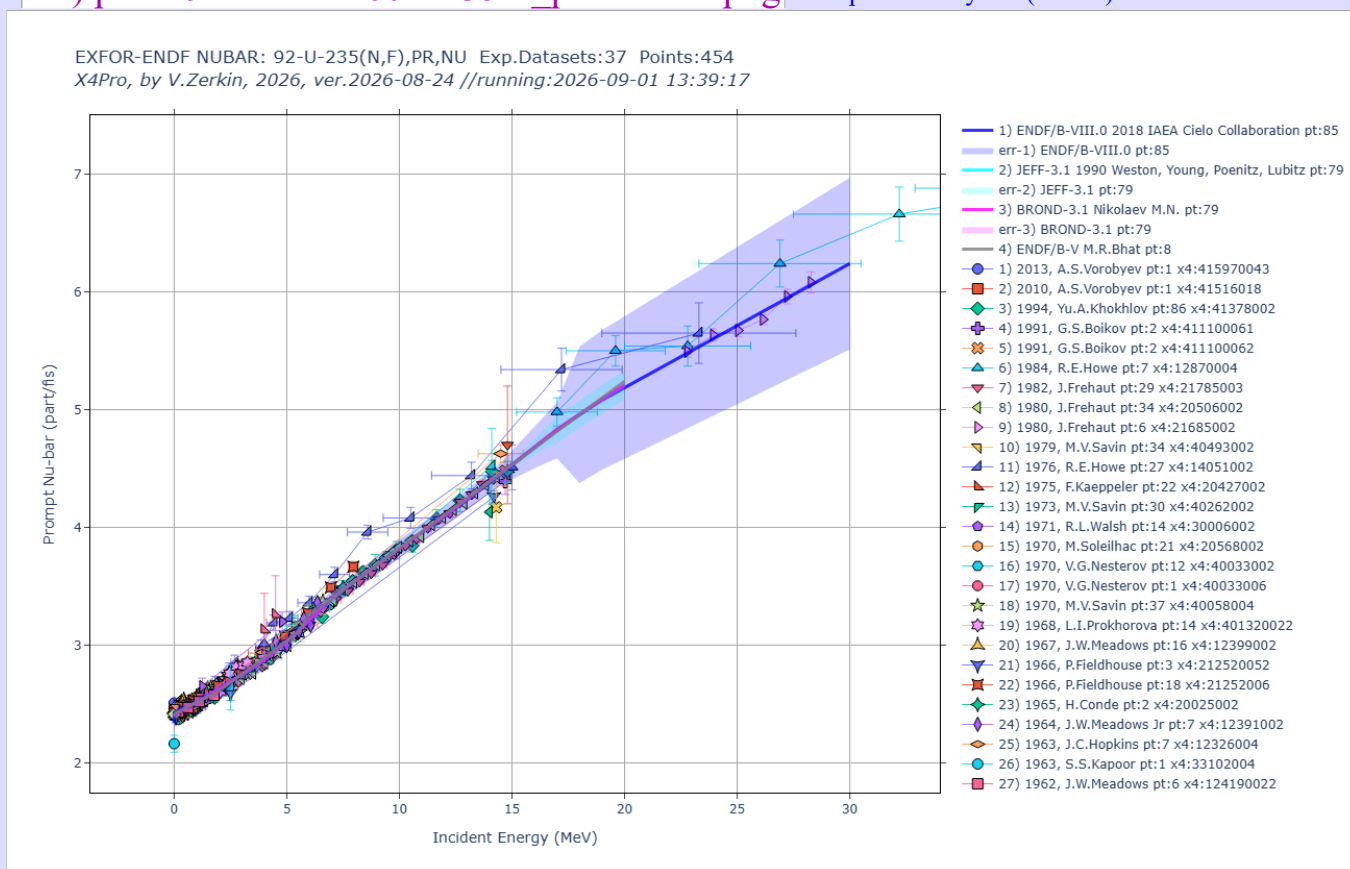


part2-9-nubar/ EXFOR-ENDF NUBAR - average number of neutrons per fission

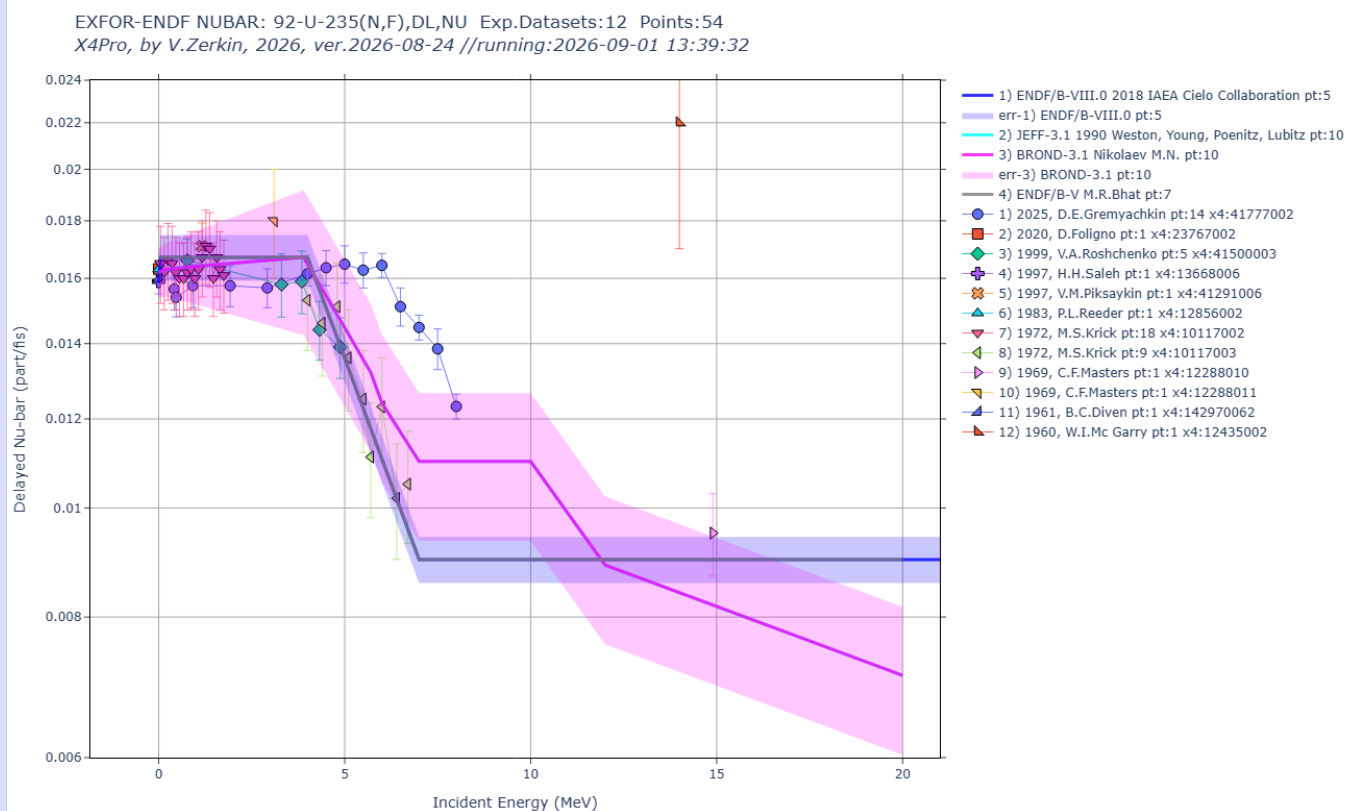
10) part2-9-nubar/out00/U235nf_-nu.html.png Total neutron yield (nu-bar)



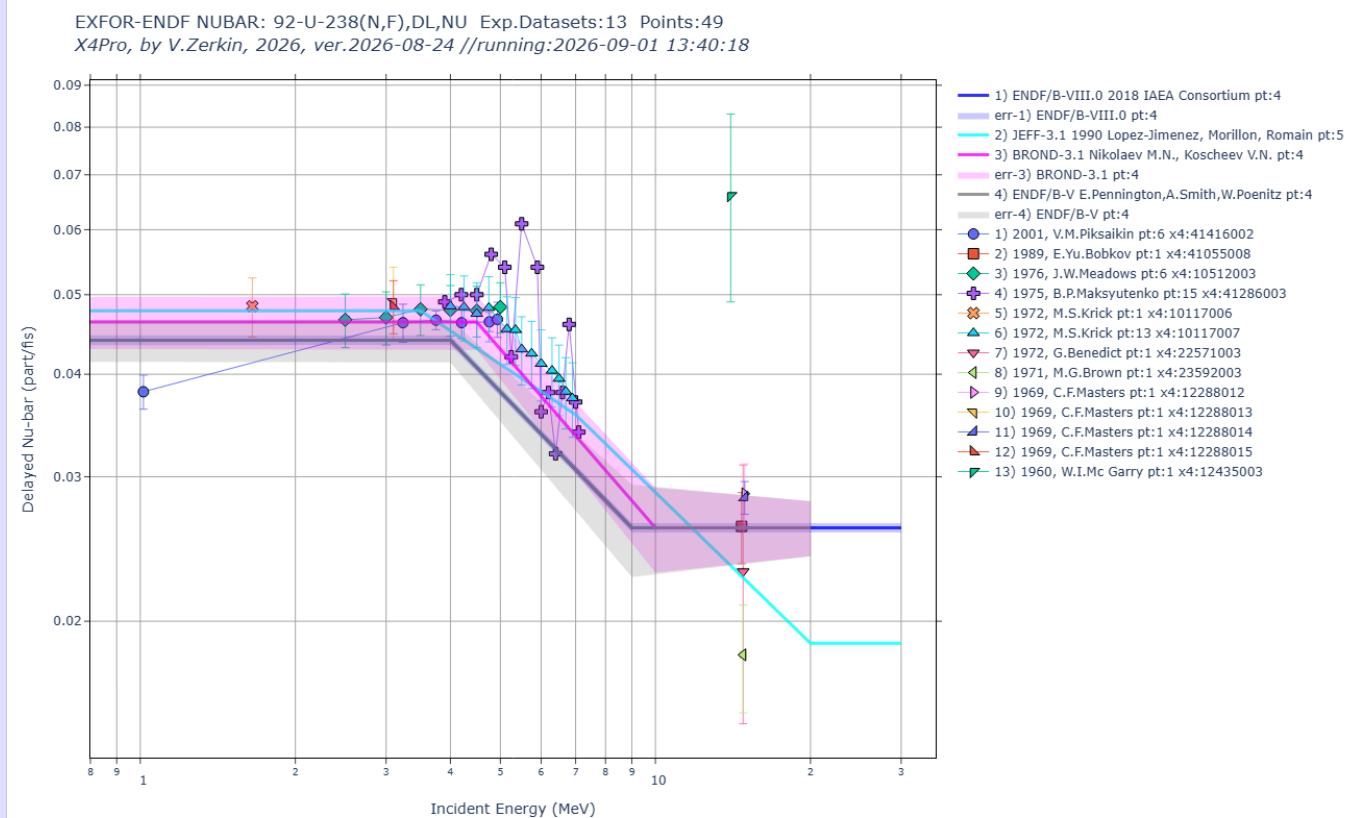
11) part2-9-nubar/out00/U235nf_pr-nu.html.png Prompt neutron yield (nu-bar)



12) part2-9-nubar/out00/U235nf_dl-nu.html.png Delayed neutron yield



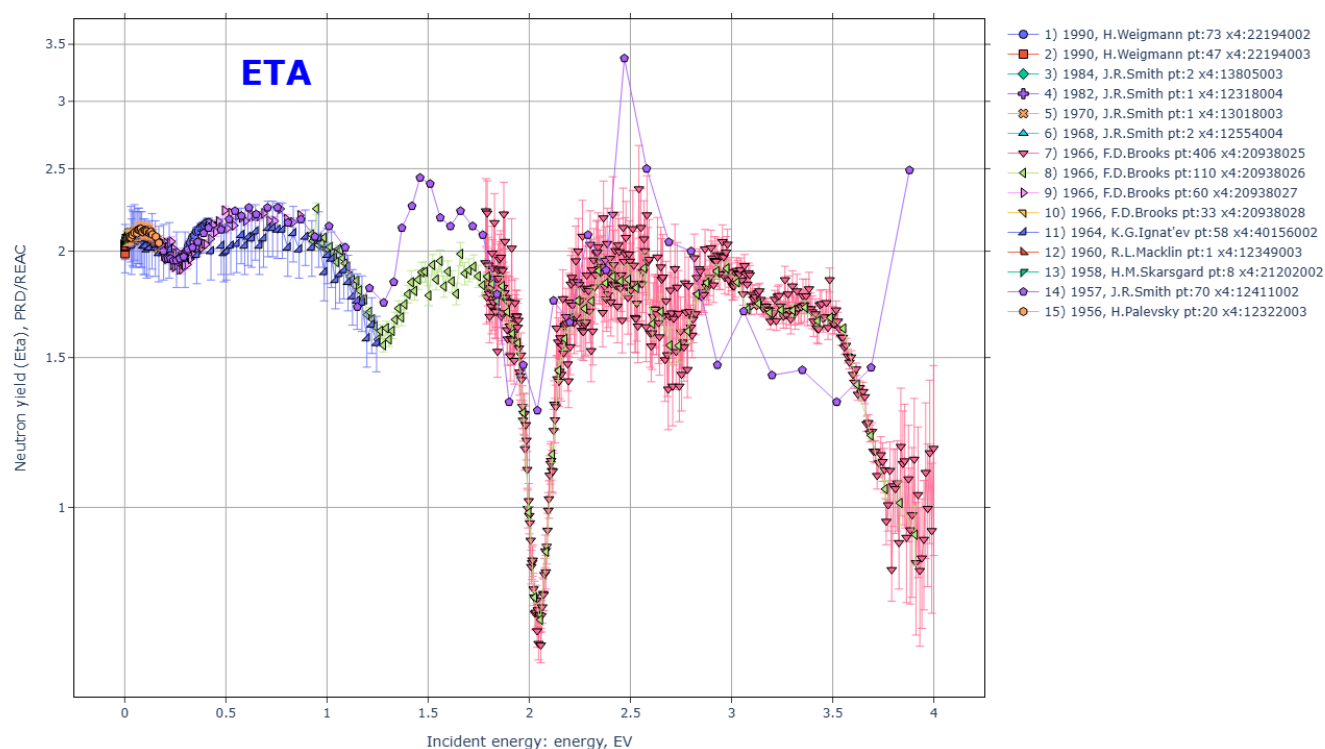
13) part2-9-nubar/out00/U238nf_dl-nu.html.png



part1-8-reac/ EXFOR universal: SIG, SIG-T, SIG-PAR, DA, DE, DA-PAR, FY, NUBAR, TKE, ETA

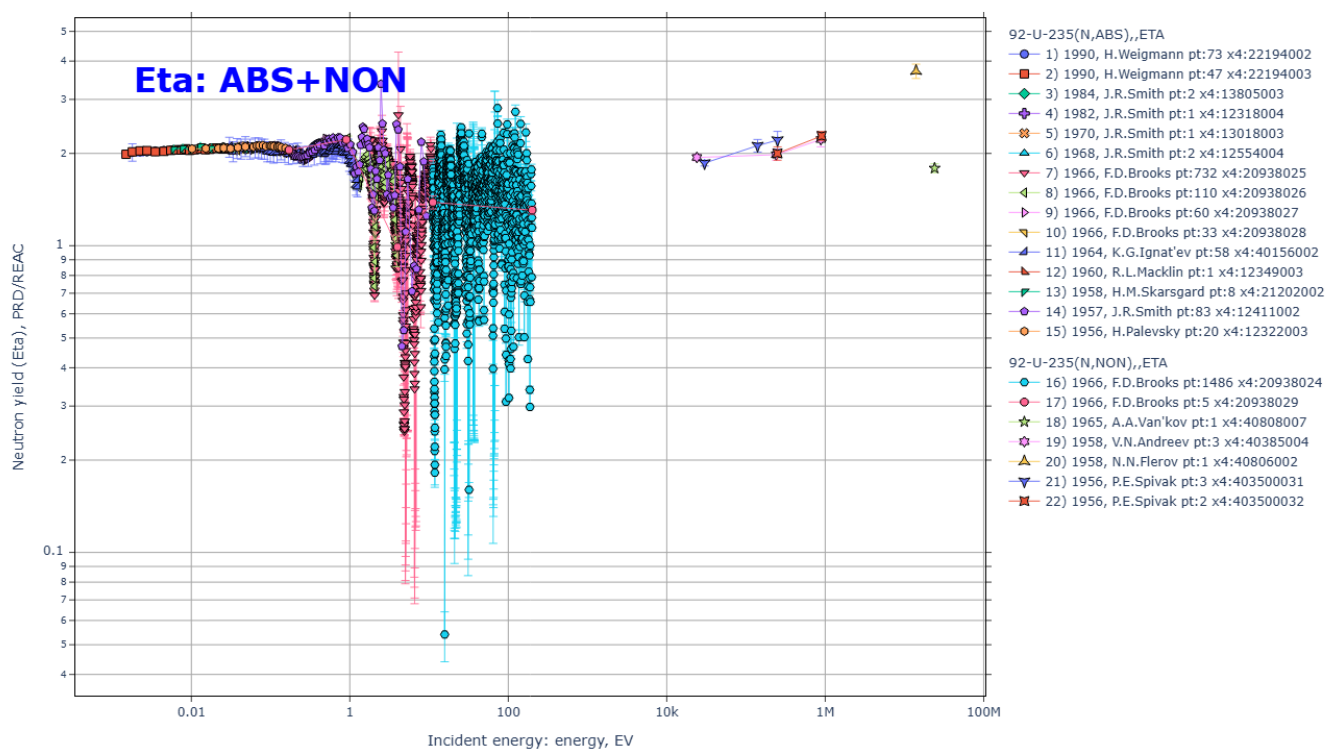
14) part1-8-reac/out00/eta1.html.png ETA Average neutron yield per nonelastic event

Reaction:92-U-235(N,ABS),,ETA Quantity:NU:y=Data(En) Datasets:15 Points:892
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-08-27 //running:2026-08-28 13:26:55

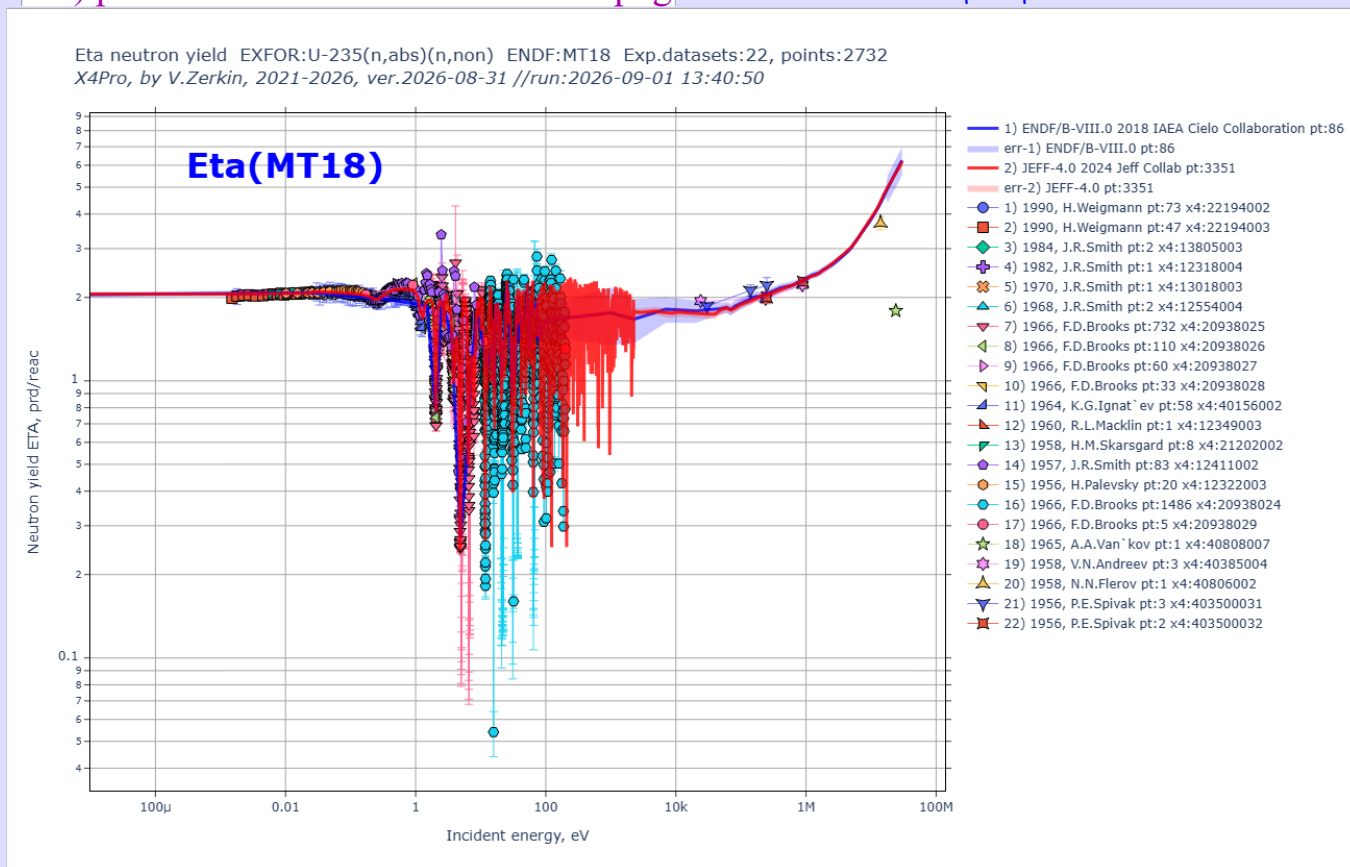


15) part1-8-reac/out00/eta2.html.png ETA: (N,ABS)+(N,NON)

Reaction:92-U-235(N,ABS),,ETA Quantity:NU:y=Data(En) Datasets:22 Points:2732
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-08-27 //running:2026-09-01 13:34:58



part2-A-eta/ EXFOR/ENDF Eta: neutron yield per nonelastic even

16) part2-A-eta/out00/U235-eta.html.png ETA: EXFOR vs. ENDF $\eta \pm \Delta\eta$ 

17) myplot1.png Interactive plot: EXFOR:ETA + ENDF:MT:452,18,102,MF:1,3,31,33

X4Pro - advanced plot
by V.Zerkin, 2025-2026